Patient ID	: QLD25/3600	Sample ID	: 2503981
Name	**: MRS B BALA**	Registration Date	: 22/03/2025 09:06 AM
Age/Gender	: 70 Yrs/Female	Report Date	: 22/03/2025 02:31 PM
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HAEMATOLOGY REPORT

Test Description	Result	Unit	Biological Reference Range
COMPLETE BLOOD PICTURE			
Hemoglobin	**10.1**	gm/dL	11.0 - 15.0
RBC Count	3.98	mil/cu.mm	3.8 - 5.5
Hematocrit HCT	**31.4**	%	36 - 48
Platelet Count	2.79	lac/cmm	1.5 - 4.5
Red Cell Indices			
Mean Corp Volume MCV	**78.9**	fL	83 - 101
Mean Corp Hb MCH	**25.4**	pg	27 - 32
Mean Corp Hb Conc MCHC	32.2	gm/dL	31.5 - 34.5
RDW-CV	12.7	%	11.6 - 14.0
Total Count and DIFFERENTIAL LEUCOCYTE COUNT			
Total WBC Count	5900	cell/cu.mm	4000 - 10000
Neutrophils	60	%	40 - 75
Lymphocytes	32	%	20 - 40
Eosinophils	02	%	01 - 06
Monocytes	06	%	02 - 10
Basophils	00	%	00 - 02
Microscopic Blood Picture			
RBC Morphology	**Anisocytosis Microcytic Hypochromic**		
WBC Morphology	With In Normal Limits		
Platelet on smear	Adequate		

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ESR - ERYTHROCYTE SEDIMENTATION RATE	**24**	mm/hr	0 - 22

Method : Automated Westergren

INTERPRETATION :

1. It indicates presence and intensity of an inflammatory process, never diagnostic of a specific disease. Changes are more significant than a single abnormal test.
2. It is a prognostic test and used to monitor the course or response to treatment of diseases like tuberculosis, bacterial endocarditis, acute rheumatic fever, rheumatoid arthritis, SLE, Hodgkins disease, temporal arteritis, polymyalgia rheumatica.
3. It is also increased in pregnancy, multiple myeloma, menstruation, and hypothyroidism.

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HBA1C/GLYCOCYLATED

HbA1c Glycosilated Haemoglobin	5.8	%	4 - 6

Method: HPLC, NGSP certified

Estimated Average Glucose :	120	mg/dL

As per American Diabetes Association (ADA)

Reference Group	HbA1c in %
Non diabetic adults >=18 years	<5.7
At risk (Prediabetes)	5.7 - 6.4
Diagnosing Diabetes	>= 6.5
Therapeutic goals for glycemic control	Age > 19 years Goal of therapy: < 7.0 Action suggested: > 8.0 Age < 19 years Goal of therapy: <7.5

ADA criteria for correlation

HbA1c(%)	Mean Plasma Glucose (mg/dL)
6	126
7	154
8	183
9	212
10	240
11	269
12	298

Note:

1. Since HbA1c reflects long term fluctuations in the blood glucose concentration, a diabetic patient who is recently under good control may still have a high concentration of HbA1c. Converse is true for a diabetic previously under good control but now poorly controlled .
2. Target goals of < 7.0 % may be beneficial in patients with short duration of diabetes, long life expectancy and no significant cardiovascular disease. In patients with significant complications of diabetes, limited life expectancy or extensive co-morbid conditions, targeting a goal of < 7.0 % may not be appropriate.

Comments: HbA1c provides an index of average blood glucose levels over the past 8 - 12 weeks and is a much better indicator of long term glycemic control as compared to blood and urinary glucose determinations.

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BIOCHEMISTRY REPORT

Test Description	Result	Unit	Biological Reference Range
KIDNEY FUNCTION TEST(KFT)			
Serum Creatinine *Method: Modified Jaffe with no deproteinzation*	1.05	mg/dL	0.60 - 1.40
Urea *Method: Urease UV GLDH*	21.27	mg/dl	10.0 - 45.0
Uric Acid *Method: Uricase Peroxidase*	2.8	mg/dl	2.5 - 6.8
Blood Urea Nitrogen-BUN *Method: Calculated*	8.5	mg/dl	7 - 20
Sodium *Method: ISE*	141	mEq/L	135 - 146
Potassium *Method: ISE*	4.0	mEq/L	3.5 - 5.5
Chloride	102	mEq/L	96.0 - 109

Interpretation :

Kidney function tests help to screen the individual for renal disease and to determine the extent or progression of renal disease. These tests also aid in determining drug dosage for drugs excreted through the kidneys. The clinical syndrome resulting from decreased renal function and azotemia is called uremia Renal azotemia: glomerular nephritis and chronic pyelonephritis. Prerenal azotemia: severe dehydration, hemorrhagic shock, and excessive protein intake. Post renal azotemia: urethral stones or tumors and prostatic obstructions Measurement of urea in dialysis fluids is widely used in assessing the adequacy of renal replacement therapy.

In these prerenal situations, the plasma creatinine concentration may be normal.In obstructive post renal conditions, both plasma creatinine and urea concentrations will be increased, although there is often a greater increase in plasma urea than creatinine because of the increased back diffusion.These considerations give rise to the principal clinical utility of plasma urea, which lies in its measurement in conjunction with that of plasma creatinine and subsequent calculation of the urea nitrogen/creatinine ratio. This ratio has been used as a crude discriminator between prerenal and postrenal azotemia. Significantly lower ratios usually denote (1) acute tubular necrosis, (2) low protein intake, (3) starvation, or (4) severe liver disease (decreased urea synthesis). So even though blood urea is not an excellent marker of renal dysfunction as it rises quite late in the dysfunction and its rise is also not exclusive to kidney dysfunction, but for practical purposes serum urea level is still one of the most ordered test and forms an important part of the kidney function test.

Concentrations in excess of 6.0 mg/dL at 32 weeks gestation have been noted to be associated with a high perinatal mortality rate.

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Fasting Plasma Glucose	97.48	mg/dl	Normal : 74 - 106
Method: Hexokinase			Impaired GT : 107 - 125
			Diabetic : >or=126

> or =126	Diabetic	Its a group of metabolic disorders characterized by a high blood sugar level over a prolonged period of time
101-125	Impaired GT	Impaired glucose tolerance means that blood glucose is raised beyond normal levels, but not high enough to warrant a diabetes diagnosis. With impaired glucose tolerance one can face a much greater risk of developing diabetes and cardiovascular disease.
70-100	Normal	The American Diabetes Association recommends a routine screening for type 2 diabetes. If the results are normal, the screening should be repeated every 3 years.

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LIPID PROFILE

Cholesterol-Total	126.86	mg/dL	< 200 - Desirable
Method: Spectrophotometry			200 - 239 -Boderline High
			> 240 - High
Triglycerides level	**270.50**	mg/dL	< 150 - Normal
Method: Serum, Enzymatic, endpoint			150 - 199 -Boderline High
			200 - 499 -High
			>500 Very -High
HDL Cholesterol	25.4	mg/dL	< 40 - Low
Method: Serum, Direct measure-PEG			> 40 - Normal
LDL Cholesterol	47.36	mg/dL	< 100 - Optimal
Method: Enzymatic selective protection			100 - 129 - Near/Above Optimal
			130 - 159 - Borderline high
			160 - 189 - High
			> 190 - Very High
VLDL Cholesterol	**54.10**	mg/dL	6 - 38
Method: Serum, Enzymatic			
CHOL/HDL RATIO	4.99		0.27 - 5.0
Method: Serum, Enzymatic			
LDL/HDL RATIO	**1.86**		2.5 - 3.5
Method: Serum, Enzymatic			

NOTE

8-10 hours fasting sample is required

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LIVER FUNCTION TEST (LFT)

TOTAL BILIRUBIN *Method: Serum, Jendrassik Grof*	0.54	mg/dl	0.1 -1.2
DIRECT BILIRUBIN *Method: Serum, Diazotization*	0.15	mg/dL	0.0 -0.3
INDIRECT BILIRUBIN *Method: Serum, Calculated*	0.39	mg/dl	0.0 -1.0
SGPT (ALT) *Method: Serum, UV with P5P, IFCC 37 degree*	16.17	U/L	0 - 49
SGOT (AST) *Method: Serum, UV with P5P, IFCC 37 degree*	35.09	U/L	0 - 46
ALKALINE PHOSPHATASE *Method: DGKC*	48.51	U/L	42 - 125
TOTAL PROTEIN *Method: Serum, Biuret, reagent blank end point*	6.20	g/dl	6.0 - 8.3
SERUM ALBUMIN *Method: Serum, Bromocresol green*	4.17	g/dl	3.4 - 5.4
SERUM GLOBULIN *Method: Serum, Calculated*	2.03	g/dl	1.8 - 3.6
A/G RATIO *Method: Serum, Calculated*	2.05		1.2 - 2.2

NOTE :

In known cases of Chronic Liver disease due to Viral Hepatitis B & C, Alcoholic liver disease or Non alcoholic fatty liver disease, Enhanced liver fibrosis (ELF) test may be used to evaluate liver fibrosis.

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Total Calcium- Serum	9.0	mg/dl	8.4 - 10.5

Method : AresenazoIII

Hypocalcemia:

Long-term therapy must be tailored to the specific disease causing the hypocalcemia. The therapeutic endpoint is to achieve a serum calcium level of 8.0 to 8.5 mg/dL to prevent tetany. For symptomatic hypocalcemia, calcium may be administered intravenously

Hypercalcemia:

The level at which hypercalcemic symptoms occur varies from patient to patient. Symptoms are common when serum calcium levels are above 11.5 mg/dL, although patients may be asymptomatic at this level. Levels above 12.0 mg/dL are considered a critical value in the Mayo Health System. Severe hypercalcemia (>15.0 mg/dL) is a medical emergency.

IRON STUDIES

SERUM IRON	41.0	ugm/dl	40 - 160
TIBC	296	ugm/dl	220 - 440
UIBC	255	ugm/dl	
Transferrin saturation	**14**	%	20 - 50

Method : Biochemical

Test Usage:

To aid in evaluation of a number of conditions involving red cell production and destruction, iron metabolism, or iron transport.

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IMMUNOASSAY REPORT

Test Description	Result	Unit	Biological Reference Range
Vitamin B12- Serum	460	pg/ml	197.0 - 771.0

Method : ECLIA

INTERPRETATION

1. Vit B12 levels are decreased in megaloblastic anemia, partial/total gastrectomy, pernicious anemia, peripheral neuropathies,chronic alcoholism, senile dementia, and treated epilepsy.
2. An associated increase in homocysteine levels is an independent risk marker for cardiovascular disease and deep vein thrombosis.
3. HoloTranscobalamin II levels are a more accurate marker of active VitB12 component.

Vitamin D- Total (25-OH-Vit D)- Serum	34.98	ng/ml	2 - 10 Defficiency 10 - 30 Insufficiency 30 - 100 Sufficiency More than - 100 Toxicity

Method : ECLIA

INTERPRETATION:

Vitamin D is a fat soluble vitamin & exists in two main forms as cholecalciferol (Vitamin D3) which is synthesized in skin from 7-dehydrocholesterol in response to sunlight exposure & Ergocalciferol (Vitamin D2) present mainly in dietary sources. Both cholecalciferol & Ergocalciferol are converted to 25 (OH) Vitamin D in liver.

Testing for 25 (OH) Vitamin D is recommended as it is the best indicator of Vitamin D nutritional status as obtained from sunlight exposure & dietary intake.

1. 25 OH Vitamin D is the best indicator of Vitamin D nutritional status, it is used as an aid in assessment of Vitamin D sufficiency in adults.
2. 25 OH Vitamin D deficiency is seen in secondary hyperparathyroidism.
3. Decreased levels of 25 OH Vitamin D can lead to Osteomalacia, reduced bone mass & thus increase the risk of bone fractures.
4. Decreased 25 OH Vitamin D levels are also associated with low bone mineral density & also seen in nutritional rickets.
5. Decreased levels of 25 OH Vitamin D are also associated with increased cardiovascular risk, low immunity & chronic renal failure.
6. Elevated levels are associated with Vitamin D intoxication.

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Thyroid Function Test (TFT)

T3	0.74	ng/mL	0.40 - 1.81
T4	9.24	µg/dL	5.00 - 12.60
TSH(Serum)	3.11	µIU/mL	0.50 - 5.50

Method : ECLIA

INTERPRETATION

TSH	T3 / FT3	T4 / FT4	Suggested Interpretation for the Thyroid Function Tests Pattern
Within Range	Decreased	Within Range	• Isolated Low T3-often seen in elderly & associated Non-Thyroidal illness. In elderly the drop in T3 level can be upto 25%.
Raised	Within Range	Within Range	•Isolated High TSHespecially in the range of 4.7 to 15 mIU/ml is commonly associated with Physiological & Biological TSH Variability. •Subclinical Autoimmune Hypothyroidism •Intermittent T4 therapy for hypothyroidism •Recovery phase after Non-Thyroidal illness"
Raised	Decreased	Decreased	•Chronic Autoimmune Thyroiditis •Post thyroidectomy,Post radioiodine •Hypothyroid phase of transient thyroiditis"
Raised or within Range	Raised	Raised or within Range	•Interfering antibodies to thyroid hormones (anti-TPO antibodies) •Intermittent T4 therapy or T4 overdose •Drug interference- Amiodarone, Heparin,Beta blockers,steroids,anti-epileptics"
Decreased	Raised or within Range	Raised or within Range	•Isolated Low TSH -especially in the range of 0.1 to 0.4 often seen in elderly & associated with Non-Thyroidal illness •Subclinical Hyperthyroidism •Thyroxine ingestion"
Decreased	Decreased	Decreased	•Central Hypothyroidism •Non-Thyroidal illness •Recent treatment for Hyperthyroidism (TSH remains suppressed)"
Decreased	Raised	Raised	•Primary Hyperthyroidism (Graves' disease),Multinodular goitre, Toxic nodule •Transient thyroiditis:Postpartum, Silent (lymphocytic), Postviral (granulomatous,subacute, DeQuervain's),Gestational thyrotoxicosis with hyperemesis gravidarum"
Decreased or within Range	Raised	Within Range	•T3 toxicosis •Non-Thyroidal illness

References: 1. Interpretation of thyroid function tests. Dayan et al. THE LANCET • Vol 357 • February 24, 2001
2. Laboratory Evaluation of Thyroid Function, Indian Thyroid Guidelines, JAPI, January 2011,vol. 59

**** End of the report. ****

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Please correlate clinically.

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